Receipt Management for Payment Wallet Applications

Minor Project -- I Report

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MASTER OF COMPUTER APPLICATION

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CONTENTS

Introduction of the project

In the modern digital economy, personal financial management is often hindered by the friction of manual data entry and the disorganization of physical and digital receipts. Individuals struggle to manage personal finances because receipts vanish quickly, manual tracking is exhausting and error-prone, and generic applications fail to provide personalized guidance. Statistics suggest that 70% of users abandon budgeting apps within 30 days due to this friction and a lack of actionable insight.

ReceiptAI is an innovative and sophisticated mobile application developed to revolutionize personal financial management by leveraging the power of Artificial Intelligence (AI) and Optical Character Recognition (OCR) technologies. The proposed solution is a cross-platform mobile application that enables users to effortlessly capture images of their physical receipts using their smartphone camera.

Once an image is captured or uploaded, the application's advanced on-device OCR engine automatically scans and extracts key transaction details—specifically the merchant name, transaction date, and total amount—thereby eliminating the need for manual data entry. Through intelligent data categorization and analysis, the app provides users with a clear and organized overview of their spending habits.

A key differentiator of ReceiptAI is its offline functionality, allowing users to scan and save receipts without an internet connection, which syncs seamlessly once connectivity is restored. Furthermore, the application integrates an AI Financial Coach powered by xAI's Grok, enabling users to ask natural language questions about their finances and receive personalized savings recommendations.

Modules identified

The development of the ReceiptAI application is divided into several distinct functional modules to ensure modularity, scalability, and efficient data processing.

2.1 User Authentication and Profile Management

This module handles user registration and secure login functionality. It ensures that user data is protected and that financial records are associated with the correct user account. It utilizes Firebase Authentication to manage credentials securely.

2.2 Receipt Capture and OCR Processing

This is the core input module of the application. It interfaces with the device hardware to capture images via the camera or select images from the gallery. It utilizes Google ML Kit for on-device Optical Character Recognition to extract text blocks corresponding to the total amount, merchant name, and date.

2.3 Data Parsing and Categorization Engine

Once text is extracted by the OCR module, the parsing engine structures this raw data. The Expense Categorization module utilizes rule-based and contextual analysis to automatically classify the transaction into predefined categories such as Groceries, Dining, Transport, and Coffee.

2.4 Data Storage and Synchronization (Offline/Online)

To support the 'Offline Mode' feature, this module manages a local on-device database for storing receipt data immediately upon scanning. A background synchronization service monitors network connectivity to sync data with the cloud-based Firebase Firestore.

2.5 Analytics and Visualization Dashboard

This module retrieves aggregated financial data and presents it to the user through an interactive UI. It generates dynamic visualizations, specifically pie charts for monthly budget distribution and bar charts for category-wise spending trends.

2.6 AI Financial Assistant (Grok Integration)

This intelligent module acts as a conversational interface. It processes natural language queries from the user and interfaces with the xAI Grok API to provide intelligent financial insights.

Flow graph of project

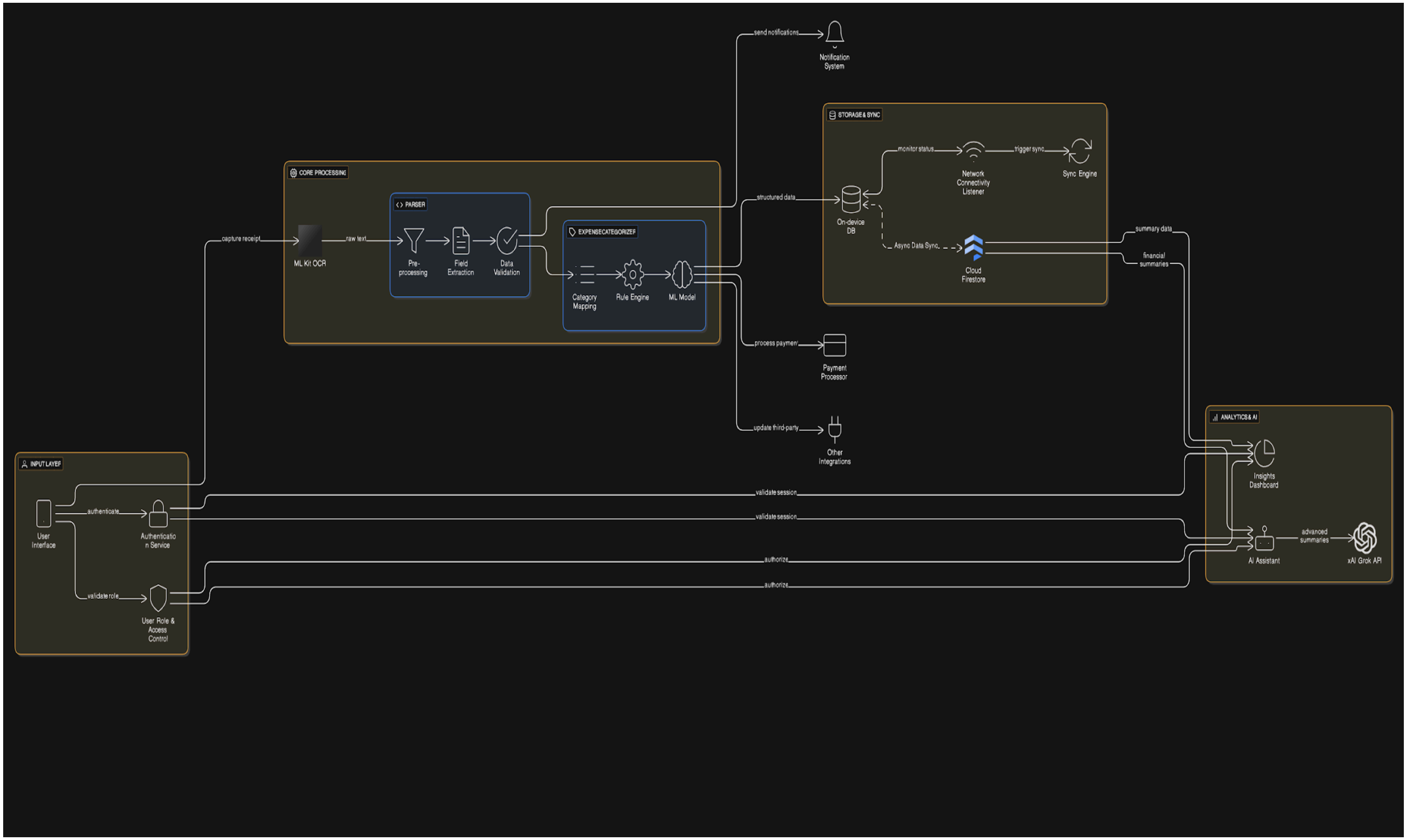
The data flow diagram (DFD) for ReceiptAI illustrates the movement of data from the user input through processing to storage and analytics.

1. Input Layer: The process begins at the User Interface, where the user initiates a Receipt Capture.

2. Core Processing: The image data is passed to the Google ML Kit OCR engine. The engine outputs raw text, which flows into the Parser. The Parser extracts structured data which is passed to the Expense Categorizer.

3. Storage Layer: The structured data is sent to the On-device DB to ensure offline capability. A synchronization service propagates this data to Cloud Firestore.

4. Analytics & AI: The Insights Dashboard fetches summary data for visualizations. The AI Assistant module fetches relevant financial summaries and interfaces with the xAI Grok API.



INPUT LAYEF
User Interface
authenticate
Authenticatio n Service
validate role
User Role & Access Control
capture receipt
CORE PROCESSING
ML Kit OCR
raw text
PARSER
Pre-processing
Field Extraction
Data Validation
EXPENSECATEGORIZEF
Category Mapping
Rule Engine
ML Model
send notifications
Notification System
structured data
process payment
Payment Processor
update third-party
Other Integrations
STORAGE & SYNC
On-device DB
monitor status
Network Connectivity Listener
trigger sync
Sync Engine
Async Data Sync
Cloud Firestore
summary data
financial summaries
validate session
validate session
authorize
authorize
ANALYTICS & AI
Insights Dashboard
AI Assistant
advanced summaries
xAI Grok API

Description of the Flow Diagram

1. Input Layer & Security

The flow initiates at the **User Interface** (mobile device), where the user interacts with the application.

- **Authentication:** Before accessing any features, the user's identity is verified via the **Authentication Service**. This module handles login credentials and session validation.
- **Access Control:** The **User Role & Access Control** layer ensures that users can only access their own private financial data, enforcing strict data privacy rules.

2. Core Processing (The Intelligence Engine)

Once a user captures a receipt, the image data undergoes a multi-stage transformation:

- **ML Kit OCR:** The raw image is processed by Google's ML Kit, which performs Optical Character Recognition to extract raw text (merchant names, prices, dates).
- **Parser:** The raw text flows into the Parser module. Here, **Pre-processing** cleans the noise, **Field Extraction** identifies specific values (e.g., "Total: $45"), and **Data Validation** ensures the numbers are logical.
- **Expense Categorizer:** The structured data is passed to the Categorization engine. Using a **Rule Engine** and **ML Model**, the transaction is automatically tagged (e.g., "Starbucks"

→→

"Coffee").

3. Storage & Synchronization (Offline-First Architecture)

This module acts as the bridge between the user's device and the cloud.

- **On-device DB:** All structured data is immediately saved to a local database. This allows the app to function perfectly without an internet connection.

- **Sync Mechanism:** A **Network Connectivity Listener** constantly monitors the device's internet status. When a connection is detected, it triggers the **Sync Engine**.
- **Cloud Firestore:** The Sync Engine performs an **Async Data Sync**, pushing local data to the cloud and pulling any updates from other devices.

4. Analytics & AI Integration

The final stage is the consumption of data for user insights.

Tools/Technologies for Implementation

4.1 Frontend Development: Flutter

The mobile application is built using Flutter, a UI toolkit developed by Google. This allows for the creation of a natively compiled application for both iOS and Android from a single codebase (Dart language).

4.2 Optical Character Recognition: Google ML Kit

For the core functionality of text extraction, Google ML Kit is employed. Specifically, the Text Recognition API is used to perform on-device OCR. This ensures low latency and enhances privacy.

4.3 Backend and Database: Firebase

Firebase is used as the Backend-as-a-Service (BaaS) solution. This includes Authentication for security, Cloud Firestore for the database, and Firebase Storage.

4.4 Artificial Intelligence: xAI Grok API

The 'Financial Coach' feature is powered by the xAI Grok API. This Large Language Model (LLM) integration allows the app to understand natural language queries.

4.5 Visualization: fl_chart

To render the financial insights, the fl_chart library for Flutter is used to create interactive pie and bar charts.

References

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